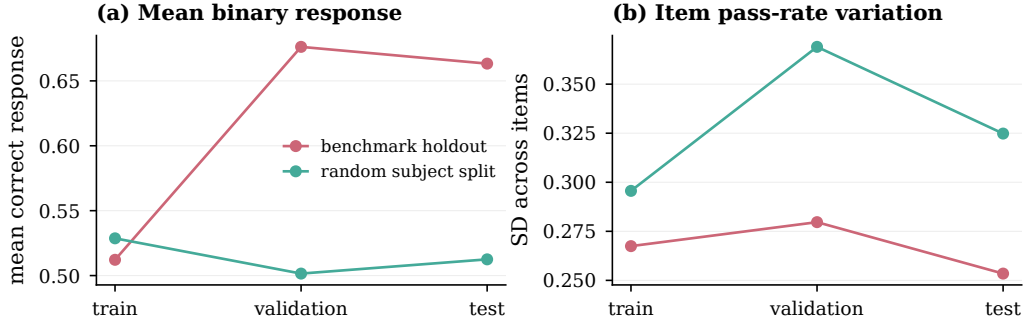
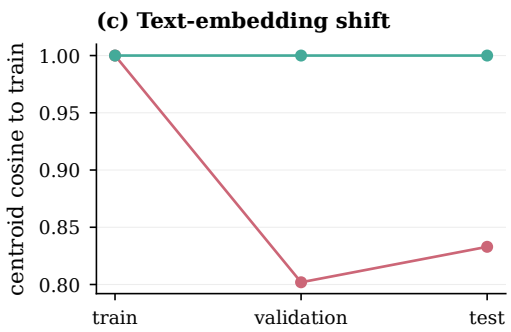
Data split diagnostics: random subject split is not benchmark transfer

Each value is descriptive. The two columns use different frozen embedding arms, so centroid cosine is interpreted within each split, not compared as an encoder ranking.





(d) Retained filtered support

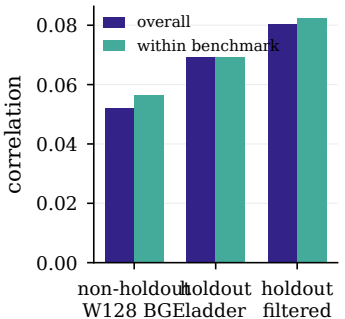
split	role	responses	tasks	items
benchmark holdo	train	5,359,699	6,013	109,987
	validation	812,782	390	3,451
	test	166,450	78	9,597
random subject	train	2,620,414	4,073	45,047
	validation	552,360	939	45,047
	test	1,174,388	1,252	56,308

Sources: docs/benchmark-holdout-split-distribution-summary.csv and docs/non-benchmark-holdout-split-distribution-summary.csv. The random-subject split shares benchmark identity with train; it is a split diagnostic, not a transfer baseline.

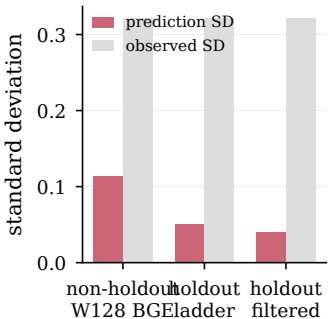
Item-difficulty transfer diagnostic

A model can adapt to labels yet still fail to predict which new items are hard. This is a descriptive 297-item held-out diagnostic, not a causal decomposition.

(a) Held-out item pass-rate correlation



(b) Predicted variation is compressed



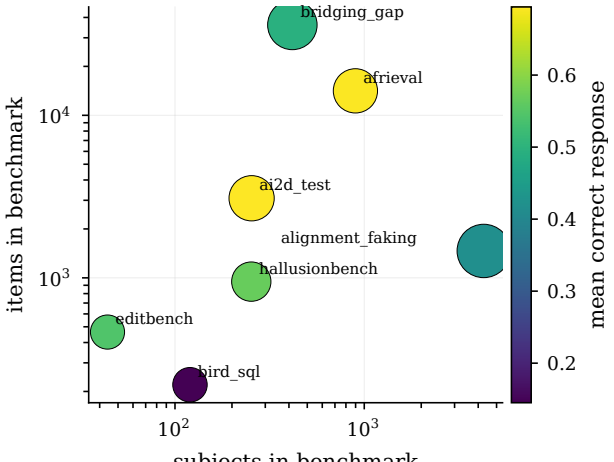
(c) Contrast in two different protocols

setting	corr	withpired 6bs SD	
non-holdout Qwe	0.339	0.277	0.109 0.325
holdout W64 filt	0.080	0.082	0.040 0.321

Sources: docs/np-online-passrate-figure11-summary.csv and docs/non-benchmark-anp128-qwen-item-passrate-summary.csv. The contrast is useful evidence for a transfer bottleneck, but differs in split, encoder, and evaluation set; do not treat it as an architecture ablation.

What is analysed in this appendix

Two different objects are kept separate: observed binary responses and frozen item-text embeddings.



Response graph

A bipartite edge joins a subject to an item only when that binary response was observed. Missing responses are not imputed.

Qwen item graph

Each filtered item connects to its 20 nearest 1024-d Qwen vectors by cosine similarity. It uses question text only.

Why separate them

Response structure needs labels and is training-only. Qwen edges can be built for a new item before its label is known.

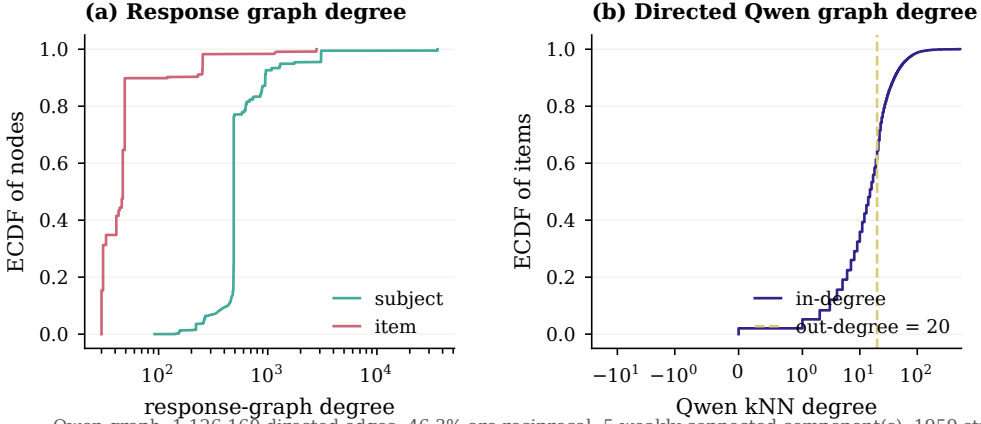
Filtering: 16,278,271 raw rows to 5,150,869; item support >= 30, subject-task responses >= 80.

How the two graphs are constructed

The response graph and the Qwen kNN graph have different edges, degree distributions, and test-time status.



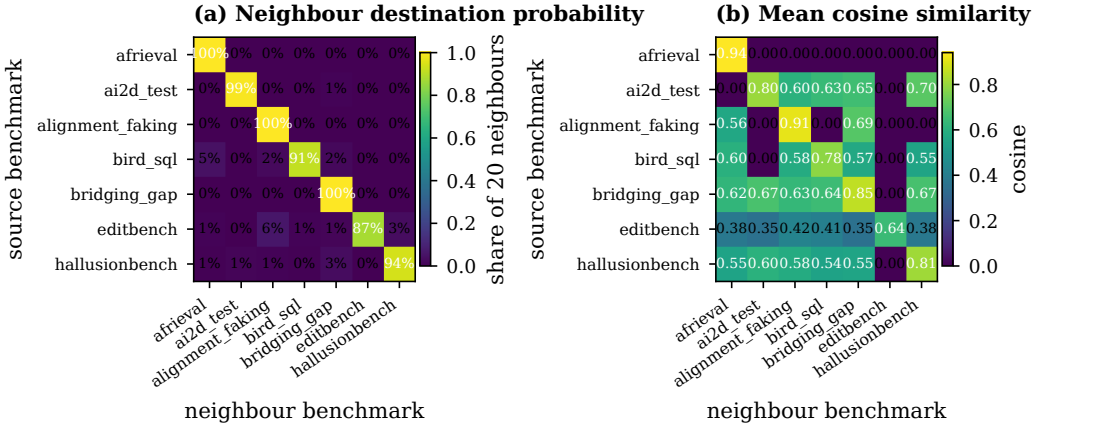
Direction comes from the source item's own Top-K list. It need not be mutual: $i \to j$ can exist while $j \not\to i$ does not.

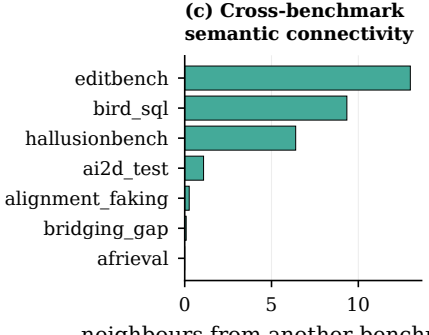


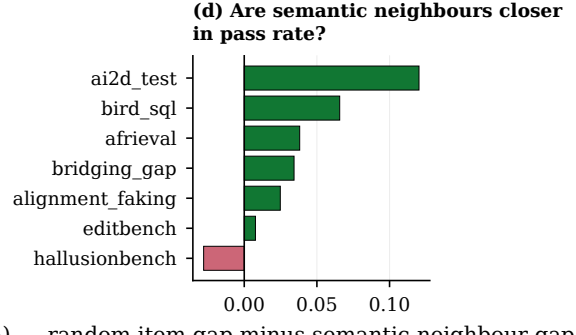
Qwen graph: 1,126,160 directed edges; 46.3% are reciprocal; 5 weakly connected component(s), 1959 strongly connected component(s).

Qwen3 item similarity forms a content-only graph

Each row contains the 20 nearest text-embedding neighbours of items from that benchmark. It is directed because each item chooses its own neighbours.



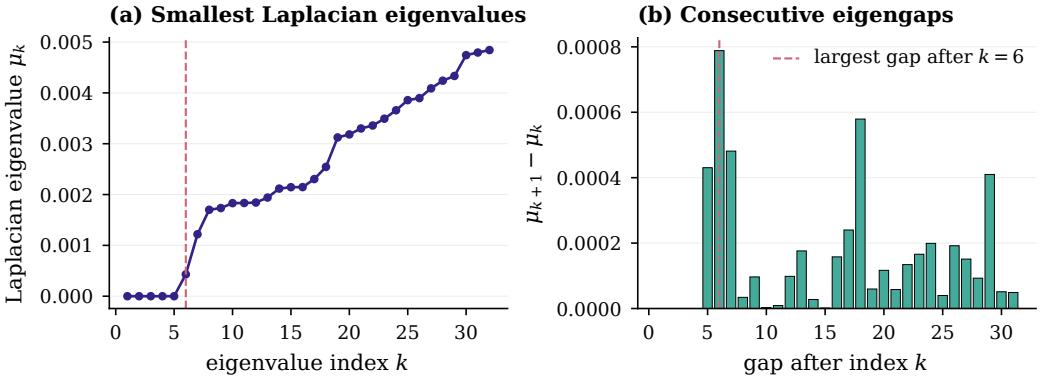




Panel (d): positive means Qwen neighbours have more similar observed pass rates than random items, after matching source and target benchmarks. It is descriptive, not a test-time score.

Laplacian spectrum of the symmetrised Qwen kNN graph

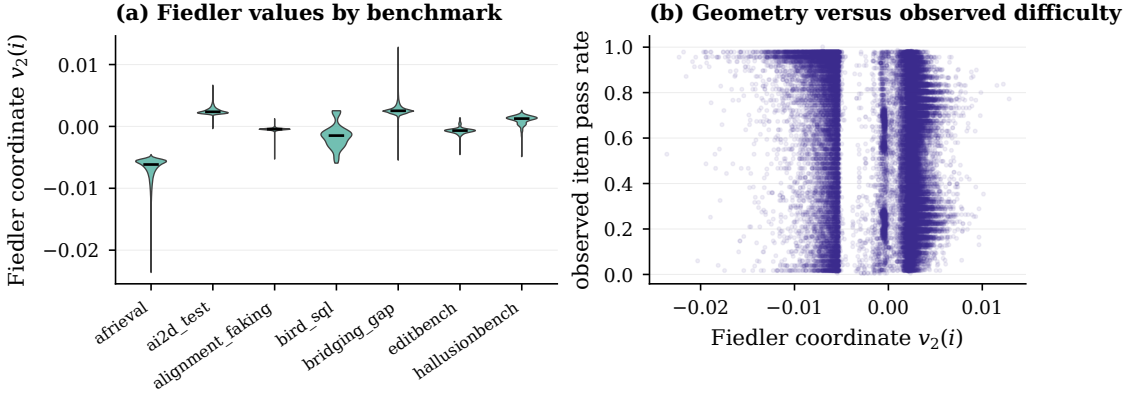
Cosine chooses each item's 20 neighbours; W is then the binary symmetrised kNN adjacency. $L_{\text{sym}} = I - D^{-1/2} W D^{-1/2}$. Near-zero values count disconnected pieces; a later eigengap suggests a coarse cluster count.



There are 5 weak components, so the first 5 eigenvalues should be near zero. The largest displayed gap is after k=6. This is a heuristic, not proof that k clusters are stable.

What the Fiedler vector can, and cannot, show

The Fiedler vector is the eigenvector for the second-smallest μ_2. It is computed inside the largest weak component because the global graph has several disconnected components and no unique global Fiedler vector.



Largest weak component: 56,221 items. The sign of v_2 is arbitrary. Separation by benchmark means the graph retains benchmark structure; a smooth relation to pass rate would suggest a difficulty-relevant semantic direction.